

Collusion-Aware Set-level Learning Framework for Detecting Spamming Activities

Yuli Liu

Quan Cheng Laboratory, Jinan & Qinghai University, Xining
China

liuyuli012@gmail.com

Abstract

Spamming activities (e.g., fake reviews, click farming, and deceptive content promotion) are increasingly conducted through collusive groups that exploit collective dynamics to manipulate platform metrics and mislead users, posing serious threats to the fairness, credibility, and functionality of online systems. To counteract these harmful behaviors, the task of spam detection has emerged as a critical area of research. However, existing detection methods generally remain limited in three key aspects: (i) They treat detection as a standard classification task, where representation learning and optimization are loosely coupled and suboptimal for capturing complex behaviors; (ii) They rely primarily on individual-level representation modeling, making it difficult to detect collective cheating strategies; (iii) They lack dedicated objective functions explicitly designed to characterize group-level spamming activities. To overcome these limitations, we introduce a collusion-aware Set-level learning framework (SetDet) that redefines the spam Detection task as a unified setwise optimization problem. Our approach offers three core advantages: (i) It enables end-to-end optimization by jointly learning representations and performing detection in a single, integrated process; (ii) It incorporates a model-level design for collusion representation, effectively capturing the temporal and relational patterns of coordinated spam; (iii) It pioneers a dedicated set-level optimization criterion that aligns closely with the structural characteristics of group-based cheating behaviors and accounts for class imbalance in real-world scenarios. Extensive experiments confirm the generalizability and superior performance of our framework across diverse spam scenarios and collusion strategies.

CCS Concepts

• Information systems → Data stream mining; Web mining.

Keywords

Spam Detection, Set-level Optimization, Sequence Modeling

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1 INTRODUCTION

Spamming behaviors are widespread across various online platforms, taking forms such as fake followers in social networks (e.g., Weibo and Twitter) [3, 47, 92], fraudulent reviews on e-commerce sites [23, 30, 64], and inflated view counts on video streaming services [9, 68]. Such deceptive behaviors are predominantly carried out through coordinated efforts [47, 48, 58], where multiple malicious entities collaborate to pursue targeted attacks, such as manipulating content rankings [32, 38], boosting commercial visibility [49, 62, 73], or misleading recommender systems [59, 88]. This collusive nature renders spamming activities more covert and evasive, as malicious users strategically mimic normal interaction patterns and distribute their actions across multiple accounts, thereby obscuring detection boundaries and significantly increasing the difficulty of identifying coordinated attacks [43, 74]. As a result, these malicious behaviors pose serious threats to the fairness, trustworthiness, and operational stability of online ecosystems.

While a substantial amount of research has explored various avenues for spam detection, the persistent evolution [4] and sophistication of deceptive tactics [56, 60] continue to present significant challenges for existing approaches: (i) The task is often approached as a classification problem. Early methods rely on feature engineering combined with traditional classification algorithms [30, 75]. More recently, the focus has shifted towards building deep learning models for representation learning, followed by a classification layer [20, 25, 80]. This decoupled paradigm lacks task-specific optimization and weakens the structural correspondence between learned representations and spamming patterns, and ultimately results in information loss and suboptimal detection performance. (ii) The vast majority of current methods are designed to identify suspicious entities at the individual level. They focus on individual-level representation learning [15, 44, 86] or feature extraction [27, 56] and ignore the coordinated and collaborative nature of cheating behaviors. As a result, these methods are often ineffective against collusive spamming strategies, where malicious entities operate in groups to evade detection mechanisms. (iii) In the field of spam detection, most existing studies still rely on general-purpose loss functions, such as cross-entropy [25, 55, 72, 77] loss and focal loss [17, 89], designed for individual-level classification tasks. This approach leads to two main issues. First, it makes it difficult to consider the collaborative relationships and dependencies among entities (spammers or targets) during the optimization process, hindering the ability to leverage group-level information. Second, it does not specifically address the inherent challenges within the spam detection domain, such as the constantly changing spam strategies [18] and the limited availability of labeled training data [12, 67] often encountered in real-world scenarios.

In response to these challenges, we propose a collusion-aware Set-level learning framework (SetDet) that formulates the spam Detection task as a unified optimization problem from a collective perspective. This framework is designed to address the challenges posed by the collusive nature of spamming activities, wherein groups of users coordinate to manipulate shared targets, thereby diluting individual-level detection signals. By jointly modeling these group-based manipulation patterns, it provides robust detection even under real-world conditions of limited labeled data [41] and varied cheating tactics [79]. Our approach is grounded in the observation that collusive spamming behaviors often manifest through coordinated user interactions with shared targets, such as groups of users collectively boosting specific products or videos. To capture such patterns, we propose a shift from the prevalent user-centric interaction paradigm [5, 10, 44, 57] in user behavior modeling to an item-centric interaction scheme. This transformation shifts the modeling perspective from isolated user behaviors to aggregated user activity patterns associated with each item, which closely aligns with the intrinsic structural properties of collusive behaviors. Building on this item-centric scheme, we introduce a dedicated collusion representation learning module that models collaborative interaction dynamics from the item's perspective. Inspired by recommendation thoughts of modeling dynamic user preferences over items from the user perspective [14, 69, 84], our module adopts an item-centric formulation that examines how groups of users collectively and temporally engage with the same target (e.g., product and video), enabling the discovery of collusive signals indicative of group-based spam manipulation.

In parallel, we pioneer a probabilistic modeling-based set-level optimization criterion tailored to address the limitations of individual-level objective functions and the need to handle the inherent complexities of spam detection. Recognizing that spamming activities often manifest as coordinated sets of actions rather than isolated incidents, SetDet optimizes the detection process at the level of related interaction sets. To further enhance the learning process and leverage the abundance of unlabeled data often available in real-world scenarios [24, 43], SetDet incorporates potentially contaminated unlabeled examples in a principled manner. This allows our method to leverage richer supervision signals and enhances its robustness. Importantly, SetDet unifies diverse forms of spamming activities under a shared item-centric interaction paradigm and supports end-to-end training, facilitating generalized collusion modeling and detection within a unified optimization framework.

Our work provides several key advantages:

- It enables a unified end-to-end set-level optimization process that tightly integrates representation learning and detection, offering a principled solution that aligns model training with the structural characteristics of collusive behaviors.
- By capturing temporal and collusive patterns directed at specific targets, the proposed item-centric collusion representation overcomes the inherent limitations of individual-level analysis to identify coordinated spam attacks.
- The tailored set-level optimization criterion based on probabilistic modeling, specifically designed to address the unique challenges of spam detection, enhances robustness against evolving spamming strategies and mitigates the impact of class imbalance.

- By introducing the generalized item-centric representation and unified optimization strategy, SetDet is broadly applicable to detecting diverse types of spam across different online platforms without requiring extensive task-specific feature engineering.

2 RELATED WORK

Spam Detection. Spam detection generally refers to the task of identifying fraudulent, manipulative, or inauthentic behaviors on digital platforms. The problem is initially studied in the context of email spam [37, 66, 76], where the goal is to classify unsolicited or malicious messages. With the evolution of online services, research expanded to opinion spam in e-commerce platforms for manipulating product reputation [30, 62], fake engagement in livestreaming platforms to boost viewer counts and popularity rankings [40, 68], and click fraud in advertising systems aimed at inflating ad revenue [22, 28]. Across these diverse spamming types, spam detection typically centers on three key components: spammers (users exhibiting fraudulent behavior), targets (e.g., products, livestreams, or advertisements), and spamming behaviors that reflects the interactions linking spammers and targets (e.g., fake reviews, artificial views, illegitimate clicks). Early approaches to spam detection rely heavily on feature engineering and traditional machine learning classifiers, trained to distinguish between benign and malicious entities [30, 54, 71]. With the rise of deep learning, recent methods have shifted toward representation learning frameworks, where user behaviors, content, and interaction patterns are embedded into high-dimensional spaces [20, 21]. A classification layer is then employed to detect spam entities [25, 80, 86].

Temporal Behavior Analysis. Temporal behavior analysis plays a vital role in understanding how users interact with systems over time [8, 19, 85]. This line of research has been widely applied across domains. For example, in web search, temporal patterns help disambiguate user intent and model session-level interactions [6, 36]; in social media, temporal signals are used to detect emerging trends or behavioral anomalies [29, 35]; and in recommender systems, temporal modeling enables the capture of evolving preferences for personalized item suggestions [33, 90]. Existing methods usually adopt a user-centric perspective [45, 65], which typically focus on modeling how individual users interact with items over time. Such approaches have shown effectiveness in tasks like recommendation [52, 69] and anomaly detection at the individual level [16, 78]. However, collusive spamming activities exhibit fundamentally different characteristics: they are characterized by coordinated user behaviors toward the specific target, involving cross-user dependencies that cannot be captured when modeling user actions in isolation. As a result, user-centric models fail to detect the coordinated nature of such spam. We therefore adopt an item-centric modeling paradigm that shifts the analytical focus from user sequences to aggregated user behaviors associated with each target.

Optimization Criterion. The optimization criterion refers to the objective function that guides the learning process of a model during training. In deep learning, the choice of optimization objective is crucial, as it directly shapes the representations learned by the model and determines how well the training aligns with the task-specific goals [50, 83]. In the recommendation domain, a wide range of task-aligned optimization criteria have been extensively explored,

such as pointwise losses for individual-level preference prediction [33], pairwise losses for ranking item relevance [26, 63], listwise objectives that directly optimize top-N ranking quality [7], and more recently, set-level formulations for capturing higher-order structural dependencies within item sets [46, 51]. In contrast, the spam detection field predominantly relies on standard classification losses, such as cross-entropy, which operate at the individual instance level [25, 55, 72, 77]. These objectives treat each entities independently, without formulating the collaborative or structural properties of group-based spamming behaviors. Despite the growing recognition of collusive behaviors in spam detection [48, 73, 87], few studies have investigated task-specific optimization criteria that explicitly reflect the group-based and structural nature of such behaviors. In particular, existing loss functions rarely incorporate the dependencies among users interacting with the same target, or account for challenges such as label sparsity [12, 13], class imbalance [2, 67], and the temporal synchronization characteristic of real-world spamming [34, 82]. This gap highlights the need for principled, structure-aware optimization objectives tailored to the detection of coordinated manipulation activities.

3 Methodology

Before presenting technical details, we first describe the conceptual differences between our formulation and conventional approaches.

3.1 Conceptual Comparison

Figure 1 outlines three key paradigms relevant to our work and highlight the distinct characteristics of SetDet. Previous approaches to spam detection, often aligned with paradigm (b), commonly adopt graph-based or interaction-based representation learning, followed by separate classification modules, typically using a classifier optimized via cross-entropy loss. In contrast, our SetDet paradigm (c) draws inspiration from temporal modeling paradigms like those in recommender systems (a) and formulates spam detection as an end-to-end optimization process without an explicit classifier. Consequently, labels in SetDet represent the status of user-item interactions in a unique way: ‘1’ denotes interactions explicitly identified as spam, ‘0’ represents labeled normal interactions, and ‘?’ indicates interactions with unknown labels during training. Furthermore, while temporal modeling in fields like recommendation (a) is often user-centric, our approach is item-centric (c). This shift is motivated by the need to model the collective activity directed at specific items, replacing the concept of ‘preference representation’ found in recommendation systems with ‘collusion representation’ to better capture coordinated deceptive behaviors. Additionally, SetDet employs a dedicated set-level optimization criterion, thus enabling simultaneous modeling of multiple related interactions. This strategy not only models richer dependency relationships among interactions but also considers the suspicious instances (unknown ones) within a set.

3.2 Preliminaries

3.2.1 Problem Formulation. As compared in Figure 1, our problem formulation fundamentally differs from previous methods. We formulate the task of detecting spamming activities as optimizing a probability distribution over potential sets of deceptive activities. Specifically, let v denote a target item, representing an object that

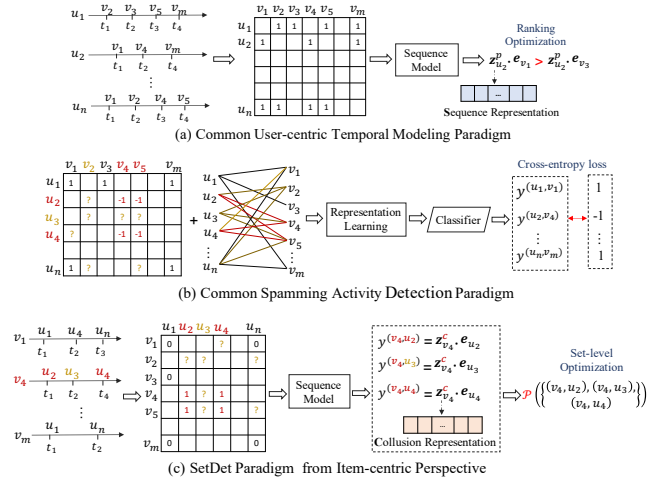


Figure 1: Comparative visualization of sequence modeling, spamming activity detection, and our SetDet framework.

spammers intend to manipulate, such as a video on a livestreaming platform, a product in an e-commerce system, or a post in a social media environment. Let U_v denote the set of users who have interacted with item v within a certain temporal window. The observed interactions between users and items correspond to spamming activities of interest, such as fake clicks on videos, fraudulent reviews on products, or coordinated likes on posts. We aim to model the probability that the set of interactions $S_v = \{(v, u) \mid u \in U_v\}$ constitutes a collusive spam set. This formulation generalizes beyond specific platforms or spamming behaviors by abstracting the detection task to a unified item-centric modeling paradigm, independent of the specific features of spamming activities.

3.2.2 Determinantal Point Processes. To operationalize the set-level optimization, we adopt Determinantal Point Processes (DPPs), a family of probabilistic models particularly suited for measuring and promoting diversity in multiple tasks, such as recommendation diversification [53, 81] and document summarization [61]. In this work, we are the first to adapt and extend the standard use of DPP from promoting diversity to explicitly capturing structural patterns of coordinated spamming behaviors. By reinterpreting DPP’s underlying mechanism, we enable it to model item-centric collusion signals, thereby systematically capturing cross-user dependencies and temporal dynamics inherent in group manipulation behaviors.

DPPs naturally capture negative correlations between entities (items), and they provide a principled mechanism to balance quality and diversity within the selected subset. DPPs are typically constructed using L-ensembles. An L-ensemble defines a DPP through a positive semi-definite matrix $L \in \mathbb{S}_+^{N \times N}$, where N is the number of items in the ground set $\mathcal{Y} = 1, 2, \dots, N$. A DPP defines a probability distribution over all subsets $Y \subseteq \mathcal{Y}$, with the probability of observing a particular subset Y given by:

$$\mathcal{P}(Y) = \frac{\det(L_Y)}{\det(L + I)} \quad (1)$$

where I denotes the $N \times N$ identity matrix, and L_Y is the principal submatrix of L indexed by the elements in Y . The determinant $\det(L_Y)$ reflects both the individual quality of items and their mutual diversity within the subset.

3.3 Collusion Representation

In sequential recommendation, the widely used Transformer models mainly rely on self-attention mechanisms to compute the relevance between interactions. However, collusive groups often exhibit globally synchronized behaviors, which are not just defined by local temporal proximity but by a set of user interactions that are strategically coordinated. To better capture these collusive signals, we propose a refined Collusive-aware Attention (CoA) mechanism. Unlike standard attention mechanisms, CoA jointly considers temporal synchronization and local interaction density, allowing the model to focus not only on the sequential closeness of user interactions but also on how concentrated and synchronized these interactions are across multiple users.

For an item v 's interaction sequence (*i.e.*, users) S^v and a maximum sequence length T , the sequence is either truncated by removing the initial interactions if $|S^v| > T$, or padded with zeros otherwise, resulting in a length-fixed sequence $\mathbf{s} = (s_1, s_2, \dots, s_T)$, where s_t represents t th user u_i that interacts with item v . A user embedding matrix $\mathbf{E} \in \mathbb{R}^{|\mathcal{U}| \times d}$ is defined for all users $\mathcal{U}$, with d representing the dimensionality. This formulation distinguishes SetDet from common recommendation models that learn user preferences over items by instead capturing the temporal and structural synchronization of multiple users toward a shared target.

In our CoA mechanism, we compute the attention distribution among interactions as follows:

$$\text{CoA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}} + \lambda \mathbf{C} \right) \mathbf{V}, \quad (2)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are the query, key, and value vectors, and $\mathbf{C}$ is the collusion affinity matrix that incorporates both temporal proximity and local interaction density, defined as:

$$C_{ij} = \exp \left(-\frac{|t_i - t_j|}{\sigma_t} \right) \cdot \frac{d_i + d_j}{2} \cdot \mathbb{I}(u_i \neq u_j), \quad (3)$$

where t_i and t_j are the interaction timestamps for users u_i and u_j . d_i and d_j are the local interaction densities around each timestamp, which measure the number of interactions in a temporal window (*e.g.*, one day). The hyperparameter σ_t controls temporal sensitivity. $\mathbb{I}(u_i \neq u_j)$ ensures that only cross-user interactions are considered. When $i = j$, the diagonal element is set to zero.

After multiple attention layers, the final hidden states $\mathbf{H}_v^{(L)}$ aggregate both temporal and collaborative signals from user interactions associated with the target item v . We finally derive a unified sequence representation $\mathbf{z}_v^c$ by applying an aggregation operation (*e.g.*, mean-pooling) over the embeddings obtained from the Transformer output. This item-centric collusion representation $\mathbf{z}_v^c$ succinctly encodes both temporal and collaborative aspects of coordinated spamming activities around item v .

3.4 Collusive Optimization for Spam Detection

In the recommendation domain, DPP-based kernels are often constructed to balance item relevance and diversity within selected sets. A typical personalized kernel matrix is formulated as:

$$\mathbf{L}^u = \text{Diag}(\widehat{\mathbf{y}}_u^S) \cdot \mathbf{K} \cdot \text{Diag}(\widehat{\mathbf{y}}_u^S), \quad (4)$$

where $\widehat{\mathbf{y}}_u^S$ denotes the calculated relevance of items [46, 50], which includes both positive (items in the interaction sequence) and randomly sampled negative items (ground set). $\mathbf{K}$ is a diversity kernel that measures pairwise similarity between items and is often constructed using a Gaussian kernel or a low-rank factorization $\mathbf{K} = \mathbf{A}\mathbf{A}^\top$, where $\mathbf{A}$ denotes trainable item representations.

To detect collusive spamming activities at the set level, we design a spam detection-specific DPP modeling method tailored to capture both individual cheating intensities and group-level collusion patterns. Specifically, we define a specialized kernel matrix $\mathbf{L}_v$ to model the internal structure of item-centric sequence. The spamming score $\widehat{y}_i^c$ for the interaction (v, u_i) is computed by taking the dot product between the user embedding $\mathbf{e}_{u_i}$ and the corresponding collusion representation $\mathbf{z}_v^c$:

$$\widehat{y}_i^c = \mathbf{e}_{u_i}^\top \mathbf{z}_v^c, \quad (5)$$

where $\mathbf{z}_v^c$ models collusive patterns of users toward item v .

To construct the item-centric collusion-aware kernel $\mathbf{L}^v$, we combine the individual spamming scores with relational information among users. The collusion similarity among users is encoded in the matrix $\mathbf{V}$, defined as:

$$\mathbf{V} = \mathbf{E}^S (\mathbf{E}^S)^\top, \quad (6)$$

where $\mathbf{E}^S$ denotes the matrix of representations for all users in the sequence S^v . Each entry $\mathbf{V}_{ij}$ measures the similarity between user u_i and user u_j , with higher values indicating stronger behavioral alignment indicative of collusion.

Using spamming score vector $\mathbf{y}^c$ for interactions in S_v and structural kernel matrix $\mathbf{V}$, we construct the spam detection-specific kernel $\mathbf{L}^v$ as:

$$\mathbf{L}^v = \text{Diag}(\widehat{\mathbf{y}}^c) \cdot \mathbf{V} \cdot \text{Diag}(\widehat{\mathbf{y}}^c). \quad (7)$$

This kernel formulation jointly encodes two aspects: (i) the spamming intensity of each interaction, and (ii) the relational structure among users reflecting their coordinated behaviors.

To optimize the detection model, we design a set-level objective based on the DPP log-likelihood. Different from traditional DPP-based recommendation frameworks that encourage diversity, in spam detection we expect coordinated users to have strong similarity rather than diversity. Therefore, for a subset $Y \subseteq S_v$, we define the set-level optimization objective as:

$$f(\Theta) = \sum_{i \in Y} \log \left((\widehat{y}_i^c)^2 \right) - \log \det(\mathbf{V}_Y^v), \quad (8)$$

where $\mathbf{L}_Y^v$ is the submatrix of user collusion representations corresponding to the interactions in Y , and Θ denotes all trainable parameters involved in computing $\widehat{\mathbf{y}}^c$ and $\mathbf{V}$. Unlike the traditional DPP probabilistic modeling, which balances relevance and diversity, SetDet introduces the '-determinant' instead of the expected '+determinant' term in the objective function. By maximizing the log-likelihood, the negative determinant term ensures that the diversity within the set is minimized, meaning that interactions within the set exhibit higher similarity. This, in turn, amplifies the collusion signals and enhances the model's ability to detect coordinated spamming activities. Through set-level optimization, our approach captures not only individual-level feature representations, but also the dependencies and collusive behaviors across the entire set. This

enables the model to detect group-level manipulation patterns, making it more effective in identifying collusive spamming activities than methods that focus solely on individual-level patterns.

3.4.1 Theoretical Analysis. To substantiate the soundness of our set-level objective and its optimization procedure, we next establish two formal results and provide their proofs.

Proposition 3.1 *For any parameter vector Θ such that every score $\hat{y}_i^c(\Theta) \neq 0$ and the collusion-specific kernel matrix $\mathbf{L}_Y^v(\Theta)$ is positive-definite, the probabilistic modeling based set-level objective is infinitely differentiable. Its gradient is*

$$\nabla_{\Theta} f(\Theta) = \sum_{i \in Y} \frac{2}{\hat{y}_i^c} \nabla_{\Theta} \hat{y}_i^c - (\mathbf{L}_Y^v)^{-1} : \nabla_{\Theta} \mathbf{L}_Y^v.$$

PROOF. Each score $\hat{y}_i^c$ equals $\mathbf{e}_{u_i}^T \mathbf{z}_v^c$. Because the embeddings $\mathbf{e}_{u_i}(\Theta)$ and $\mathbf{z}_v(\Theta)$ are smooth functions of Θ , the score is smooth as well. The scalar function $x \mapsto \log x^2$ is analytic on $x \neq 0$; therefore the first summation in f is infinitely differentiable. The collusion matrix is defined as $\mathbf{L}^v = \mathbf{E}^S (\mathbf{E}^S)^T$. Since $\mathbf{E}^S(\Theta)$ depends smoothly on Θ and yields a positive-definite matrix, the composition with the analytic function $-\log \det(\cdot)$ is smooth. Consequently f is C^∞ .

Differentiating $\log x^2$ gives $2/x$. For a positive-definite matrix $\mathbf{M}$, the derivative of $-\log \det \mathbf{M}$ with respect to $\mathbf{M}$ is $-\mathbf{M}^{-1}$. Applying the chain rule to both parts of f leads immediately to the stated gradient expression. $\square$

Proposition 3.2 *Assume that the gradient of f is L -Lipschitz continuous on a compact parameter domain containing the initial point Θ_0 . Gradient descent with fixed step size $\eta \in (0, 1/L]$, that is*

$$\Theta_{k+1} = \Theta_k - \eta \nabla f(\Theta_k),$$

produces a sequence satisfying $\|\nabla f(\Theta_k)\| \rightarrow 0$. Thus the iterates converge to a first-order stationary point.

PROOF. Lipschitz continuity of ∇f implies the descent inequality

$$f(\Theta_{k+1}) \leq f(\Theta_k) - \frac{\eta}{2} \|\nabla f(\Theta_k)\|^2.$$

for any $\eta \leq 1/L$. Because f is bounded below on the compact domain, $f(\Theta_k)$ is monotonically decreasing and convergent. Summing the descent inequality over k yields a finite bound on $\sum_k \|\nabla f(\Theta_k)\|^2$, which forces the gradient norm to vanish in the limit. Hence gradient descent converges to a first-order stationary point. $\square$

The theoretical analysis above confirms that the set-level objective is fully differentiable with an explicit gradient and that standard first-order optimizers converge reliably, validating the practicality of our SetDet framework.

3.4.2 Complexity Analysis. In the Transformer encoder, the projection of queries, keys, and values takes $O(Td^2)$, the attention weight computation requires $O(T^2d)$, and the output projection costs another $O(Td^2)$. Thus, each layer has complexity $O((Td^2 + T^2d))$. For the set-level optimization, evaluating $\log \det(\mathbf{L}_Y)$ and its gradient has a worst-case cost of $O(|\mathcal{G}|^3)$, where $|\mathcal{G}|$ denotes the size of the ground set for a given item-centric sequence. Therefore, the overall time complexity is $O(Td^2 + T^2d + |\mathcal{G}|^3)$. In practice, the ground set size $|\mathcal{G}|$ (i.e., the number of candidate interactions involved in set-level evaluation) is typically much smaller than the full sequence

length T , allowing the complexity to be simplified as $O(T^2d)$ when $d > T$ and $|\mathcal{G}| \ll T$.

In terms of space complexity, the model stores Transformer parameters of size $O(d^2)$, embedding tables of size $O(|\mathcal{U}|d + |\mathcal{V}|d)$ ($|\mathcal{U}|$ and $|\mathcal{V}|$ denote user and item sets, respectively) and maintains $O(Td)$ space per sequence for hidden states and projections. Attention weights require $O(T^2)$ space per layer. Combining these, the total space complexity is $O(d^2 + Td + T^2 + |\mathcal{U}|d + |\mathcal{V}|d)$, which is dominated by $O(d^2 + T^2)$ assuming large user and item vocabularies are stored externally. Since T is fixed (e.g., $T = 20$) in all experiments, both time and space complexity remain tractable and suitable for real-world deployment. Therefore, overall, our method maintains a tractable computational cost per item-centric sequence, with both time and space complexity scaling quadratically with the sequence length and the embedding dimension, making it well-suited for deployment in real-world spam detection scenarios.

4 Experiments

To evaluate the effectiveness of SetDet in detecting spamming activities, we conduct a series of experiments on real-world datasets with diverse spamming scenarios on different platforms. The **implementation details** can be found in Section A.

4.1 Experimental Settings

4.1.1 Datasets. The empirical validation of our spam detection framework employs three datasets representing fundamentally diverse spamming phenomena on different online platforms:

- For relationship manipulation on microblogging platforms, we leverage the **Social** dataset in [47]. It comprises 3K verified fake follower accounts obtained through crowdsourced underground market transactions. Item-centric interaction matrix construction adopts a bipartite paradigm where followees serve as target nodes (i.e., the items that spammers aim to manipulate) attracting following relationships from follower accounts (user nodes). Following an evidence-based edge labeling protocol, we identify spam links (spamming activities) as connections between purchased fake followers and followees that exhibits abnormal follower acquisition patterns (≥ 2 fake followers), thereby simulating real-world coordinated following attacks.
- The **Forum** dataset from [11] addresses promotion campaigns detection in discussion forums. Originating from a documented deceptive campaign (Samsung incident), this dataset encapsulates 148K user replies with temporal metadata, mapped onto an item-centric interaction network where initial posts constitute target nodes and responder accounts form user nodes. Spam edges are explicitly defined as replies posted by 312 confirmed spammers (0.52%), creating a realistic evaluation scenario for detecting low-prevalence deceptive content dissemination.
- To evaluate the effectiveness of our method in e-commerce scenarios, we adopt the **Amazon** Movies dataset provided by [88] for the task of detecting spamming product reviews (opinion spam). To ensure data quality, we filter out cold-start users and retain only those with at least 100 historical reviews, resulting in a refined dataset containing 297,073 reviews from 5,519 active users across 70,633 movies. Potential spammers are identified based on community-driven quality signals: users whose reviews receive fewer than 30% “helpful” votes are flagged for manual

inspection, through which 792 accounts are confirmed as malicious. An item (*i.e.*, a movie) is labeled as a malicious target if it is reviewed by more than one verified spammer. Conversely, normal users are defined as those whose review helpfulness ratio exceeds 50%, and products with no known spammer engagement are considered legitimate. We construct a movie-user interaction matrix where each interaction corresponds to a review, labeled as a spamming activity if it originates from a confirmed spammer and targets a malicious item; otherwise, it is labeled as non-spam.

4.1.2 Compared Methods. We compare 13 representative baselines to demonstrate the superiority of our SetDet framework:

- **Label Propagation-Based Methods.** We include two classical label propagation methods originally designed for node-level spam detection, focusing on identifying either spammers or manipulated targets. Specifically, Graph_{HITS} [47] is inspired by HITS-like iterative propagation for detecting suspicious nodes in interaction graphs. Graph_{Bi} [88] refines the common bipartite graph propagation approach by applying reweighted edges. As these methods are inherently designed to assign spam scores to nodes, not edges, we adapt them to our interaction-level detection setting by computing the average of the predicted spam scores from both endpoints of each edge (*i.e.*, user and item).
- **Feature-Based Models.** These models adopt logistic regression classifiers trained on hand-crafted features commonly used in spam detection literature. For the Social dataset, we extract relational features for fake following relationship detection, following the methodologies proposed in [3] and [47], referred to as FeatureSocial. For the Forum and Amazon datasets, we employ content- and behavior-based features designed for opinion spam detection, as introduced by [11] and [31], denoted as Featurecontent. These feature sets serve as strong traditional baselines representing prior domain-specific knowledge.
- **Structure-Aware Learning Models.** These methods integrate both graph structure and node-level features to enhance spamming activity detection. Graph_{Factor} is a probabilistic factor graph model proposed by [48], which jointly models correlation and attribute factors to capture both relational and behavioral evidence of spamming. We also include Graph_{GCN} [39], a semi-supervised approach based on GCNs, which propagates both structural and feature-based signals across the graph to infer node labels. Both models represent advanced graph-based learning paradigms.
- **Latent Factor-Based Models.** To emphasize the uniqueness of SetDet in modeling collusive behavior, we compare with three representative latent factor models that capture structural or preference patterns in user-target interactions. Latent_{SVD} [1, 88] applies Singular Value Decomposition (SVD) to the raw user-target interaction matrix without considering sign information. It relies solely on latent structural signals extracted from the interaction graph for spam detection. Latent_{GAUC} [70] focuses on optimizing the top vs. bottom positions of a predicted ranking list under a signed network setting. These methods were originally proposed for recommendation tasks. In our setting, we repurpose the learned latent factors as input features to a classifier for spamming activity detection. Latent_{n-aware} [42] introduces a negative-aware matrix factorization approach that explicitly models the uncertainty of unobserved user-item interactions.

- **Recent Proposed Baselines.** We further compare against several recently proposed competitive models for spam detection. SplitGNN [80] leverages spectral graph neural networks to enhance detection via graph frequency analysis. GFD [91] improves performance by applying a partitioned message-passing mechanism, which reduces information noise and enhances representation learning within local graph regions. LFM-SPR [51] introduces a signed latent factor model to learn user and item embeddings, using signed relations to detect spamming behaviors in bipartite graphs. Feature_{free} [43] is a contrastive node-level spam detection model that avoids handcrafted features by learning pairwise signals to assign higher spam scores to suspicious nodes. For edge-level evaluation, we compute interaction scores by averaging the predicted values of the connected nodes.
- **Pruned SetDet Models.** To validate the core idea of our method, *i.e.*, detecting spamming activities based on collusive set-level ranking optimization criterion, we perform an ablation study by constructing three simplified variants of SetDet: (i) SetDet_{-C} removes CoA by replacing it with a standard attention module (*i.e.*, pruning the collusion affinity matrix); (ii) SetDet_{-Set} replaces our proposed set-level DPP-based optimization with a conventional individual-level cross-entropy loss; (iii) SetDet_{-both} removes both the collusion-aware attention and the set-level optimization components. This ablation study also helps demonstrate the impact of each component on the overall performance of SetDet.

4.2 Overall Comparison

Table 1 reports the comparison results between SetDet (including variants) and competitive baselines spanning different technical paradigms. Several important observations can be drawn:

- SetDet consistently outperforms all baselines across all datasets. This demonstrates the effectiveness of our framework in capturing both temporal spamming signals and group-level collusion behaviors through end-to-end set-level ranking optimization.
- The improvements are substantial across different platforms and spamming activity phenomena, including fake following on social media, opinion spam in e-commerce and promotion campaigns in online forums, highlighting the generalizability and applicability of SetDet in handling diverse real-world scenarios involving various types of coordinated spamming activities.
- The superiority of SetDet holds under different proportions of labeled data. Even with limited supervision, SetDet achieves strong performance, indicating its robustness in real-world conditions where labeled spamming and normal activities are often sparse. This also confirms SetDet’s ability to address the challenge of label imbalance commonly encountered in spam detection.
- The ablation variant SetDet_{-C}, which removes the collusion affinity matrix and uses standard attention, shows only a minor drop in performance compared to the full SetDet. This suggests that while the collusion affinity contributes to learning refined temporal alignment, the backbone collusion representation and collusive optimization remains robust.
- The variant SetDet_{-Set}, which removes the set-level optimization and uses individual-level cross-entropy loss, resulting in a more significant performance drop compared to SetDet_{-C}. This indicates that set-level modeling plays a critical role in enhancing the framework’s ability to capture collusive behaviors. Notably,

Table 1: AUC performances of compared methods on three datasets with varying training set size (%). The best results are in bold, and the second-best (excluding SetDet variants) are underlined. SetDet improvements are statistically significant ($p < 0.05$).

Methods	Social			Forum			Amazon		
	30%	40%	50%	30%	40%	50%	30%	40%	50%
Graph _{HITS}	0.762	0.775	0.782	0.671	0.680	0.687	0.700	0.718	0.752
Graph _{Bi}	0.774	0.783	0.791	0.679	0.685	0.690	0.708	0.730	0.757
Feature _{Social}	0.703	0.729	0.742	-	-	-	-	-	-
Feature _{Content}	-	-	-	0.563	0.581	0.595	0.670	0.687	0.699
Latent _{SVD}	0.742	0.769	0.783	0.645	0.664	0.671	0.705	0.734	0.760
Latent _{GAUC}	0.775	0.798	0.819	0.697	0.710	0.735	0.736	0.758	0.772
Latent _{n-aware}	0.785	0.804	0.821	0.698	0.736	0.740	0.741	0.760	0.775
Graph _{Factor}	0.782	0.796	0.806	0.692	0.702	0.710	0.745	0.756	0.805
Graph _{GCN}	0.824	0.850	0.865	0.731	0.747	0.776	0.784	0.813	0.829
Feature _{free}	0.792	0.808	0.826	0.704	0.720	0.739	0.754	0.770	0.803
SplitGNN	0.820	0.847	0.860	0.722	0.752	0.769	0.780	0.806	0.840
GFD	0.827	0.861	0.868	0.733	0.757	0.773	0.805	0.817	0.833
LFM-SPR	0.859	0.864	0.871	0.752	0.764	0.771	0.800	0.820	0.838
SetDet _{-C}	0.869	0.881	0.890	0.770	0.776	0.784	0.813	0.838	0.850
SetDet _{-Set}	0.855	0.870	0.865	0.758	0.767	0.775	0.809	0.815	0.830
SetDet _{-both}	0.847	0.861	0.862	0.754	0.762	0.770	0.796	0.808	0.825
SetDet	0.878	0.885	0.897	0.775	0.784	0.796	0.833	0.851	0.869

even without the probabilistic DPP-based objective, SetDet still outperforms all baselines, demonstrating that the item-centric collusion representation itself is highly effective. This further confirms the strength of SetDet’s design: collusion-aware modeling and item-centric structure jointly enables a unified set-level optimization process, demonstrating that its overall design is not only principled but also well-aligned with the structural nature of coordinated spamming activities.

- When the proportion of labeled training instances is low (e.g., 30%), the performance gap between SetDet and baselines becomes more pronounced. This demonstrates SetDet’s strength in handling real-world challenges such as limited supervision and class imbalance. This advantage stems from the set-level optimization design, which allows selectively incorporating unlabeled but suspicious interactions into the constructed ground set. For example, given an item v_i with three labeled spam interactions and two unlabeled ones, SetDet can stochastically sample one unlabeled interaction into a spam-likely subset Y of size 4. Since elements in Y are labeled as spam or suspicious ones, the optimization still favors collusion indicative patterns, enabling the model to exploit unlabeled instances in a principled and effective manner.

4.3 Properties Analysis

To further understand the effectiveness and underlying characteristics of SetDet, we conduct additional property analyses beyond overall performance. In Equation 3, σ_t controls the sensitivity to temporal proximity. We empirically evaluate several settings and find that $\sigma_t = 1$ hour consistently yields the best overall performance across all three datasets. This setting effectively captures the typical temporal patterns of collusive spamming behaviors, which tend to occur in loosely coordinated bursts over extended periods, rather than at narrowly aligned timestamps. All results reported in Table 1 are obtained with $\sigma_t = 1$ hour.

To better evaluate the effectiveness of SetDet in identifying spamming activities, we present Precision@K results under varying values of K . As shown in Figure 2, SetDet consistently outperforms

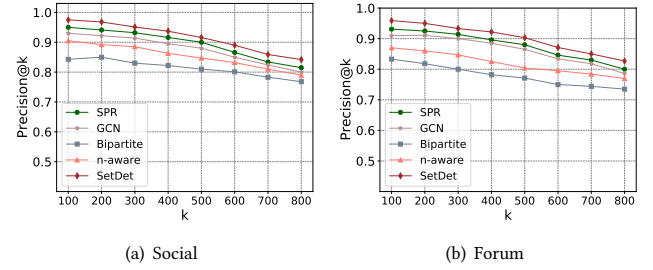


Figure 2: Precision@K performance on different datasets.

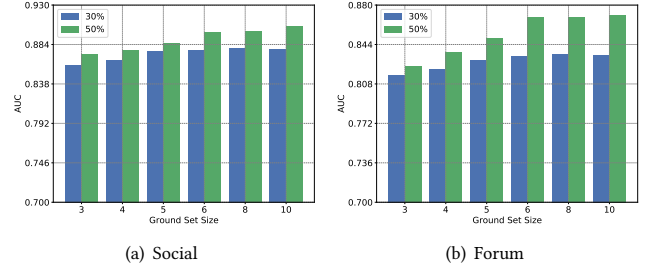


Figure 3: Performance w.r.t. ground set size of DPP modeling.

several representative baselines across all K values. Notably, the improvement is more pronounced at smaller K , indicating that SetDet is especially effective at ranking true spamming interactions higher. This reflects the strength of our collusion-aware set-level modeling in accurately capturing suspicious patterns and ranking them ahead of benign behaviors. By ensuring that high-risk spamming activities are prioritized, SetDet enables more efficient and targeted intervention, making it particularly suitable for practical deployment in security-sensitive platforms.

We next examine the effect of the DPP ground set size $|\mathcal{Y}|$ (i.e., the number of candidate interactions sampled per item for optimization) on detection performance. As shown in Figure 3, we report AUC results on the two datasets as $|\mathcal{Y}|$ increases from 3 to

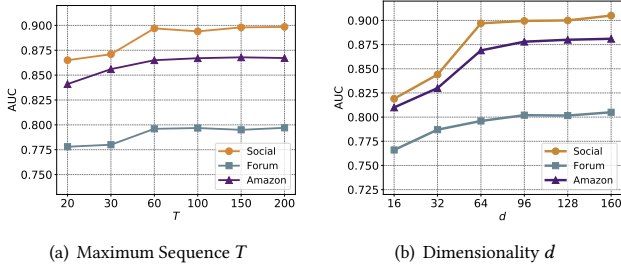


Figure 4: Impact of parameters T and d on different datasets.

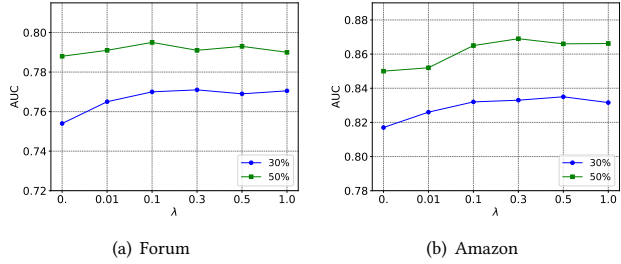


Figure 5: Performance w.r.t. λ on different datasets.

10. The performance trend reveals two clear phases. Initially, as $|\mathcal{Y}|$ grows from small values, the model benefits significantly: a larger ground set allows the DPP-based optimization to capture richer dependency structures among interactions, thereby better modeling collusive patterns. This confirms the importance of considering inter-instance relationships, rather than treating activities as isolated. However, once the ground set size reaches around 6, the performance gains begin to diminish. This plateau suggests that most informative collusive signals are already captured, and further increasing the set size introduces redundant interactions that offer limited additional value. These results collectively demonstrate that SetDet is capable of leveraging a moderate ground set to effectively model collusion, without requiring exhaustive interaction coverage. In all experiments, we fix $|\mathcal{Y}| = 6$, as it offers a strong and stable detection performance. This choice reflects a principled design decision, rather than a trade-off made by sacrificing training efficiency in pursuit of marginal performance improvements.

Figure 4 reports performance of SetDet while varying the maximum sequence length T (from 20 to 200) and the embedding dimension d (from 16 to 160) on three datasets. The curves exhibit a consistent trend: performance improves as T or d increases and then saturates once moderate values are reached. A larger T allows the model to consider more historical interactions for each item, thereby capturing longer-range collusion cues; likewise, a larger d offers additional expressive capacity to encode subtle temporal and structural patterns. It is noteworthy that throughout all reported experiments we simply fix $T = 60$ and $d = 64$ (values commonly adopted in sequential modeling literature) for every dataset, without any per-dataset hyper-parameter tuning. The fact that SetDet achieves state-of-the-art results under this unified configuration demonstrates its robustness and ease of deployment.

We further analyze the impact of parameter λ , which controls the contribution of the collusion affinity matrix in our CoA Attention mechanism. As shown in Figure 5, when $\lambda = 0$, the collusion

affinity is disabled, resulting in lower detection performance across all datasets. Performance notably improves as λ increases from 0, indicating the benefit of explicitly modeling collusion patterns. However, excessively large λ values lead to performance degradation, suggesting that overemphasis on temporal synchronization can overshadow other meaningful interaction signals. In all reported experiments (Table 1), we adopt $\lambda = 0.3$, which consistently yields optimal and stable detection performance.

We further evaluate SetDet in terms of robustness to evolving spamming strategies and training efficiency. To simulate evolving spamming behaviors, we randomly inject 3% additional interactions associated with labeled spamming targets, mimicking coordinated attempts to conceal malicious activities by adding seemingly normal interactions. In comparative experiments across the three datasets (with 50% training set size), competitive baselines SplitGNN and SPR exhibit average performance drops of 4.5% and 3.9%, respectively, while SetDet’s performance only decreases by 2.6%. This highlights SetDet’s superior robustness against evolving collusive strategies due to its effective modeling of set-level collusion patterns rather than relying solely on fixed feature representations. Moreover, when setting the ground set size $|\mathcal{Y}| = 6$, SetDet demonstrates training times comparable to representative deep-learning-based baselines (SplitGNN, GFD, and SPR), with less than a 14% increase in runtime across all three datasets. Given that SetDet consistently achieves significantly higher detection performance than these methods, this result underscores our model’s favorable balance between computational efficiency and detection effectiveness.

5 Conclusion

In this work, we present SetDet, a novel probabilistic modeling based framework for spam detection that introduces a set-level optimization criterion tailored, specifically designed to capture the complexities of collusive spamming activities. Unlike traditional methods that focus on individual user-level interactions, SetDet adopts an item-centric representation, enabling it to model and capture coordinated behaviors across groups of users targeting shared items. SetDet leverages the power of Collusive Multi-head Attention (CoA) to enhance its ability to model collusion patterns within an item sequence. Our experimental results demonstrate the superior performance of SetDet across multiple real-world datasets, outperforming 13 competitive baselines. The model not only achieves high accuracy but also exhibits robustness in handling evolving spamming activities, which are often characterized by dynamic and deceptive behavior. The set-level optimization and item-centric design contribute to SetDet’s strong generalizability, making it applicable to diverse spamming scenarios across different platforms. By introducing a novel approach to spam detection, SetDet addresses key challenges such as label imbalance and limited supervision, providing a more scalable and adaptive solution. Overall, our work establishes a new paradigm in spam detection, where collaborative behaviors and temporal relationships are explicitly modeled to improve detection performance and robustness.

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A Implementation Details

Our proposed SetDet framework is implemented using PyTorch, and experiments are conducted on a NVIDIA RTX A5000 GPU. To fine-tune the hyperparameters, we perform an extensive grid search across all compared methods, and report performance based on the peak validation results. The embedding dimension d for representation learning method is tested for values in the set $\{32, 64, 128\}$, while the maximum sequence length T is varied within $\{20, 50, 100, 200\}$. The learning rate is optimized within the range $\{10^{-3}, 10^{-4}\}$. We also explore dropout rates in the set $\{0.1, 0.3, 0.6\}$ and adjust λ

within $\{0, 0.01, 0.1, 0.3, 0.5, 1.0\}$. For Transformer backbone, we investigate the number of layers ($\{1, 2, 3\}$), the number of attention heads ($\{1, 2, 4\}$). σ_t of SetDet is used to control the temporal sensitivity in the collusion affinity computation, determining how strongly the model emphasizes time synchronization among interactions. We validate σ_t with values $\{10, 30, 60, 120\}$ (in minutes). To validate the applicability and stability of our method, we vary the proportion of randomly selected labeled spamming activities used in the training set. Specifically, we test with training set sizes of 30%, 40%, and 50%, with the remaining data evenly divided into validation and test datasets. To ensure a fair comparison, all methods are evaluated using the same data splits.